

Industrial Internship Report on " Banking Information System"

Prepared by

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Executive Summary

This report presents the work completed during the **4-week Core Java Internship** organized by **Upskill Campus, The IoT Academy**, in collaboration with **UniConverge Technologies Pvt. Ltd. (UCT)**.

During this internship, I developed a **Banking Information System**, a console-based application using **Core Java**. The project enables users to register new accounts, securely log in, deposit and withdraw money, check account balances, transfer funds, view transaction history, change passwords, and store account information using file handling.

The project was developed using **Object-Oriented Programming (OOP)** concepts such as classes, objects, encapsulation, and methods. File handling was implemented to save and retrieve account details, ensuring data persistence.

This internship enhanced my practical knowledge of Core Java, strengthened my programming and problem-solving skills, and provided valuable experience in developing a real-world application. It also improved my understanding of software development practices and GitHub version control.

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1 Preface

This internship provided me with an excellent opportunity to gain practical knowledge of **Core Java** and understand how programming concepts are applied in real-world applications. During this **4-week internship** organized by **Upskill Campus, The IoT Academy, and UniConverge Technologies Pvt. Ltd. (UCT)**, I worked on developing a **Banking Information System**.

The main objective of this project was to design and develop a console-based banking application that allows users to perform essential banking operations such as account registration, secure login, deposit, withdrawal, balance enquiry, fund transfer, transaction history, account details, and password management. The project also uses **file handling** to store account information permanently.

Throughout the internship, I strengthened my understanding of **Object-Oriented Programming (OOP)** concepts including classes, objects, encapsulation, methods, and constructors. I also improved my skills in **ArrayList, file handling, debugging, logical thinking, and project development** using Java.

This internship has increased my confidence in developing Java applications and has given me valuable experience that will be helpful for my academic projects and future career in software development.

I sincerely thank **Upskill Campus, The IoT Academy, UniConverge Technologies Pvt. Ltd. (UCT), Vishnu Institute of Technology**, and all the mentors and faculty members for their continuous guidance and support throughout this internship.

Finally, I encourage my juniors and fellow students to actively participate in internships and complete every task sincerely, as practical experience plays an important role in improving technical skills and career development.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies** e.g. **Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.



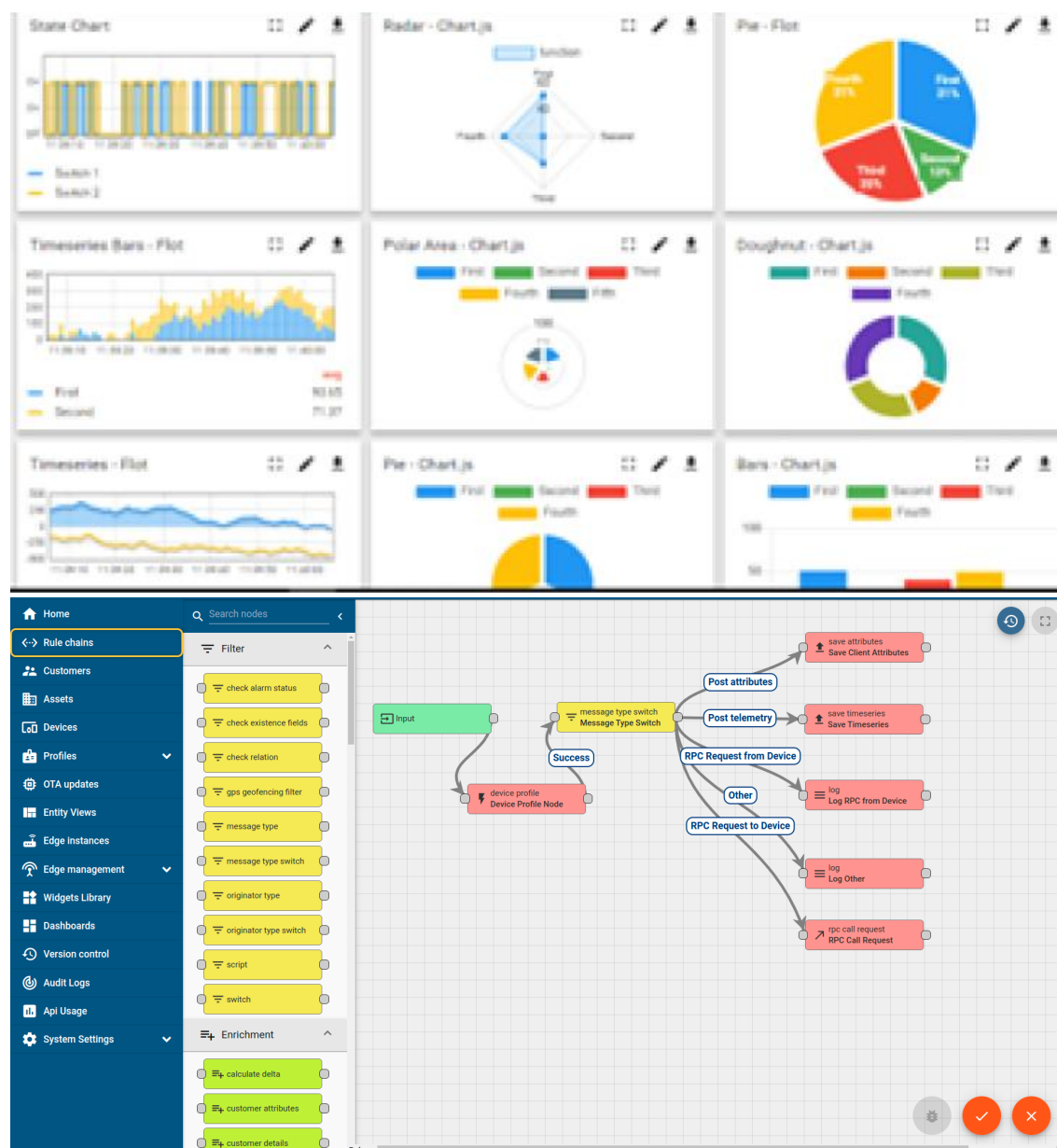
i. UCT IoT Platform ()

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine



FACTORY WATCH

ii. Smart Factory Platform ()

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleash the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they want to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.



Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
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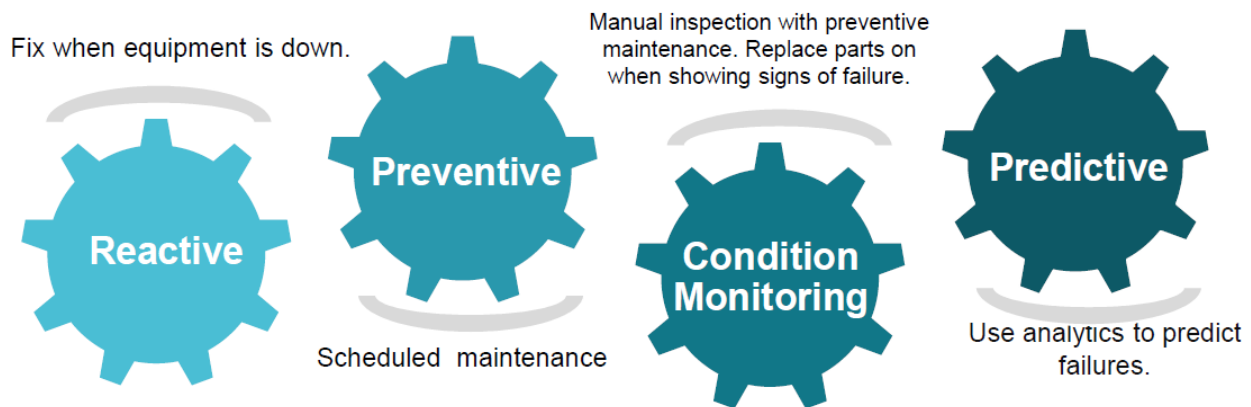


iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN technology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

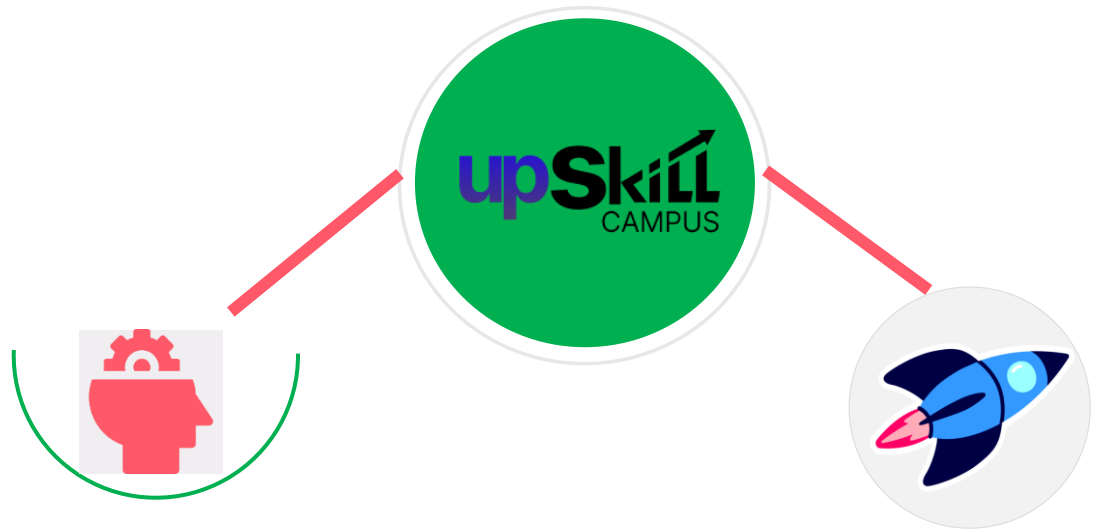
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

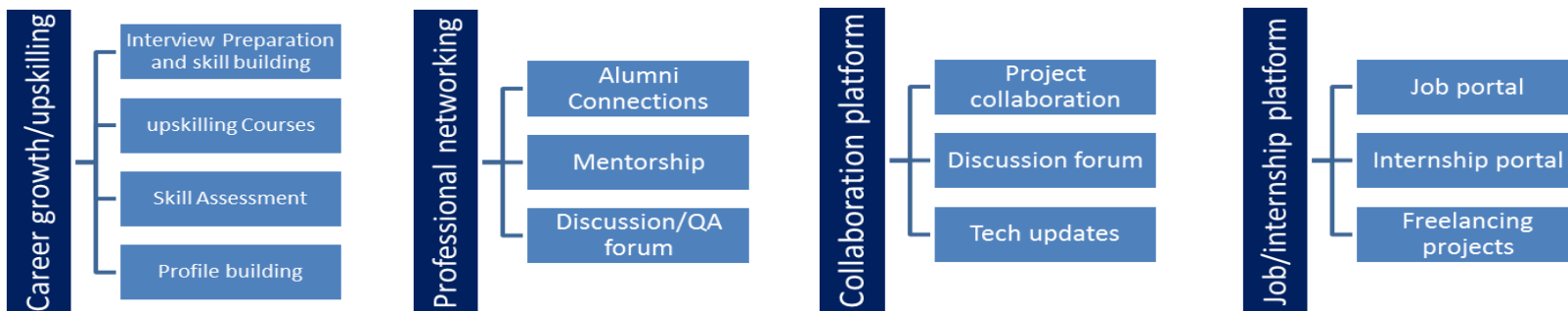
USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

<https://www.upskillcampus.com/>



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

- get practical experience of working in the industry.
- to solve real world problems.
- to have improved job prospects.
- to have Improved understanding of our field and its applications.
- to have Personal growth like better communication and problem solving.

2.5 Reference

- [1] Oracle Java Documentation – <https://docs.oracle.com/javase/>
- [2] GitHub Repository – <https://github.com/Kiransai217/Banking-Information-System>
- [3] Upskill Campus Learning Portal
- [4] Course materials provided during the internship.

2.6 Glossary

Terms	Acronym
OOP	Object-Oriented Programming
JVM	Java Virtual Machine
JDK	Java Development Kit
JRE	Java Runtime Environment
IDE	Integrated Development Environment

3 Problem Statement

In today's world, banking services require secure and efficient management of customer accounts and financial transactions. Maintaining these records manually is time-consuming, prone to errors, and makes it difficult to manage customer information effectively.

The objective of this project is to develop a **Banking Information System** using **Core Java** that allows users to perform basic banking operations through a console-based application. The system provides features such as:

- User Registration
- Secure Login using Account Number and Password
- Deposit Money
- Withdraw Money
- Balance Enquiry
- Transaction History
- Fund Transfer
- Account Details
- Change Password
- Data Storage using File Handling

The application is developed using **Object-Oriented Programming (OOP)** concepts such as classes, objects, encapsulation, and methods. It also uses **ArrayList** to manage account data during execution and **file handling** to save and retrieve account information from a file, ensuring data persistence.

This project helps users perform essential banking operations efficiently while improving programming skills in Java and providing practical experience in developing a real-world application.

4 Existing and Proposed solution

4.1.1.1 Existing System

The existing system mainly relies on manual record keeping or basic applications that have several limitations. Manual banking operations require more time, increase the chances of human error, and make it difficult to manage customer information efficiently. Many simple applications also do not provide secure login, transaction history, or permanent data storage.

Limitations of the Existing System:

- Time-consuming manual record management.
 - Higher possibility of human errors.
 - Lack of secure user authentication.
 - No proper transaction history.
 - Difficulty in maintaining customer records.
 - Limited data storage and retrieval.
-

5 Proposed Solution

The proposed solution is a **Banking Information System** developed using **Core Java**. The application provides a secure and user-friendly console interface for performing basic banking operations. It uses **Object-Oriented Programming (OOP)** principles and **file handling** to store account details permanently.

Features of the Proposed System:

- User Registration
 - Secure Login
 - Deposit Money
 - Withdraw Money
 - Balance Enquiry
 - Transaction History
-

- Fund Transfer
- Account Details
- Change Password
- Persistent Data Storage using File Handling (accounts.txt)

5.1 Code submission (Github link)

Repository:

<https://github.com/Kiransai217/Banking-Information-System>

5.2 Report submission (Github link) : first make placeholder, copy the link.

6 Proposed Design/ Model

The **Banking Information System** is a console-based application developed using **Core Java**. The system follows a simple workflow where users can register an account, log in securely, and perform various banking operations. The application uses **Object-Oriented Programming (OOP)** concepts and **file handling** to manage and store account information.

6.1 High Level Diagram (if applicable)

Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

6.2 Low Level Diagram (if applicable)

6.3 Interfaces (if applicable)

The project provides a console-based user interface with the following options:

Main Menu

- Register
- Login
- Exit

User Menu

- Deposit
- Withdraw
- Check Balance
- Transaction History
- Transfer Funds
- Account Details
- Change Password
- Logout

The interface is simple, user-friendly, and menu-driven, allowing users to perform banking operations efficiently through the command line.

7 Performance Test

The Banking Information System was tested to ensure that all major functionalities work correctly and efficiently. Different test cases were executed using valid and invalid inputs to verify the system's reliability, accuracy, and data storage. The application successfully performed all banking operations without errors and stored account information using file handling.

7.1 Test Plan/ Test Cases

Test Case	Input	Expected Result	Status
Register Account	Valid user details	Account created successfully	Pass
Login	Correct Account No. & Password	Login successful	Pass
Login	Invalid Password	Login failed	Pass
Deposit	Valid amount	Balance updated	Pass
Withdraw	Amount less than balance	Withdrawal successful	Pass
Withdraw	Amount greater than balance	Insufficient balance message	Pass
Transfer Funds	Valid receiver account	Money transferred successfully	Pass
Change Password	Correct old password	Password updated	Pass
View Transaction History	Existing transactions	Transaction history displayed	Pass

7.2 Test Procedure

- Launch the application.
- Register a new bank account.
- Log in using the generated account number and password.
- Perform deposit and withdrawal operations.
- Check the updated account balance.
- Transfer funds to another account.
- View transaction history.
- Change the account password.
- Log out and log in again to verify the updated password.

- Restart the application and verify that account details are successfully loaded from the `accounts.txt` file.

7.3 Performance Outcome

The Banking Information System successfully completed all test cases. The application handled user authentication, account management, deposits, withdrawals, fund transfers, and transaction history correctly. File handling ensured that account details were saved and loaded properly, making the system reliable and suitable for small-scale banking operations. The project met all the objectives defined at the beginning of the internship.

8 My learnings

The 4-week Core Java Internship was a valuable learning experience that helped me improve both my technical and problem-solving skills. Through the development of the **Banking Information System**, I gained practical knowledge of Java programming and understood how software applications are designed and implemented.

During this internship, I learned the following:

- Improved my understanding of **Core Java** programming.
- Applied **Object-Oriented Programming (OOP)** concepts such as classes, objects, encapsulation, constructors, and methods.
- Learned to use **ArrayList** for efficient data management.
- Implemented **File Handling** to store and retrieve account information permanently.
- Developed a menu-driven **console application** with multiple banking features.
- Improved my logical thinking, debugging, and problem-solving abilities.
- Learned to organize Java projects and maintain source code using **GitHub**.
- Understood the importance of software documentation and project reporting.

This internship increased my confidence in developing Java applications and provided practical experience that will help me in future academic projects, internships, and campus placements. It also motivated me to continue learning advanced Java technologies and software development concepts.

9 Future work scope

The **Banking Information System** developed during this internship successfully performs the basic banking operations using Core Java. However, there are several enhancements that can be implemented in the future to make the application more secure, efficient, and user-friendly.

Some of the future improvements are:

- Develop a **Graphical User Interface (GUI)** using Java Swing or JavaFX for a better user experience.
- Replace file handling with a **MySQL database** to manage customer records more efficiently.
- Implement **OTP or email-based authentication** for enhanced account security.
- Add an **Admin Module** to manage customer accounts and monitor transactions.
- Introduce features such as **mini statements, loan management, and fixed deposit services**.
- Enable **online banking features** such as bill payments, mobile recharges, and fund transfers through different banks.
- Generate and download **account statements** in PDF format.
- Improve the application's security by encrypting user passwords and sensitive information.
- Develop a **mobile application** version to allow users to access banking services from smartphones.

10 Conclusion

The Banking Information System project helped me understand how Core Java concepts can be applied to develop a real-world application. Through this internship, I gained practical experience in Object-Oriented Programming, file handling, debugging, and project development. This project strengthened my programming skills and increased my confidence in developing software applications. The knowledge and experience gained during this internship will be valuable for my future academic projects, internships, and career in software development.